

PPOL 5205: Data Science III - Advanced Modelling Techniques
Insights Report
Music Recommendation System Based on Spotify Audio Features
Muhammad Saad - ms4689

Summary

I developed a content-based music recommendation system using Spotify audio features, which explored how songs can be recommended based on their audio characteristics rather than on user behavior. The main aim was to develop a recommendation system that can help users with music discovery.

Research Objective

Music recommendation systems help users discover and consume music on streaming platforms. Most music recommendation systems, including the ones currently in use on platforms such as Spotify and YouTube, rely on collaborative filtering. This approach does not consider the audio features of a song. Instead, it focuses on user behavior and listening history to recommend songs as part of a network. Although collaborative filtering is useful, it has its limitations. Since songs are recommended purely on the basis of other users' preference, it is possible that only popular songs receive visibility.

Recent advances in Music Information Retrieval (MIR) have now made it possible to quantitatively encode the audio features of songs (Schedl, Gómez, & Urbano, 2014). These features can be used to develop machine learning algorithms that recommend songs based on similarity of user preferences (Scaringella, Zoia, and Mlynek, 2006; Kathavate, 2021). This content based approach solely relies on the audio features of the songs themselves, instead of user interaction or engagement.

I wanted to leverage the content-based approach to develop a music recommendation system using a database of Spotify songs that contained audio features. The goal was to understand whether songs can be meaningfully grouped using audio attributes and whether these groupings can be used to generate intuitive recommendations.

Data

I used a publicly available dataset titled the '*Spotify Tracks Dataset*'. It is hosted on Kaggle and has been compiled by Maharshi Pandya (2021), and is accessible [here](#). The data was extracted by the Kaggle user through the Spotify API while it was still available to researchers (it is only available to registered and paying app developers now). It had a total of 21 variables and 114,000 songs, which included the audio features of songs, metadata, and engagement metrics.

I only focused on audio features, which included, along with their short description:

1. *Danceability*: Measure of how suitable a song is for dancing
2. *Energy*: Measure of a song's intensity, including how fast or noisy it is
3. *Valence*: Measure of how positive a song is
4. *Acousticness*: Measure of the extent to which a song uses non-electronic instruments
5. *Speechiness*: Measure of the extent of spoken words or lyrics in a song
6. *Instrumentalness*: Measure which can be intuitively thought of as opposite to speechiness and captures the lack of vocals in a song

7. *Liveness*: Measure of whether and to what extent was a song recorded in front of a live audience
8. *Loudness*: Measure of the perceived volume in a song
9. *Tempo*: Measure of the speed at which music is played in a song

Based on the description above, it seemed that some of these features were closely related to one another. For example, Speechiness and Instrumentalness seem the exact opposite of each other. Similarly, Energy and Loudness seem to overlap. Figure 1 shows the correlation between audio features to investigate this. It showed that some audio features were indeed highly correlated, such as Energy and Loudness, Valence and Danceability, and Acousticness and Energy.

In order to get a more intuitive understanding of how audio features captured genres, I analyzed the top genres by the highest average for audio features such as Acousticness, Energy and Danceability. The top three genres by the highest average Acousticness in the dataset were: classical, romance and tango; by Energy: death metal, grind core and happy; by Danceability: kids, house music and reggae. Some of these are captured in Figure 2.

Since the aim was to develop a music recommendation system that did not rely on collaborative filtering, metadata related to songs such as genre, popularity as well as Spotify track ID was not used in the algorithm.

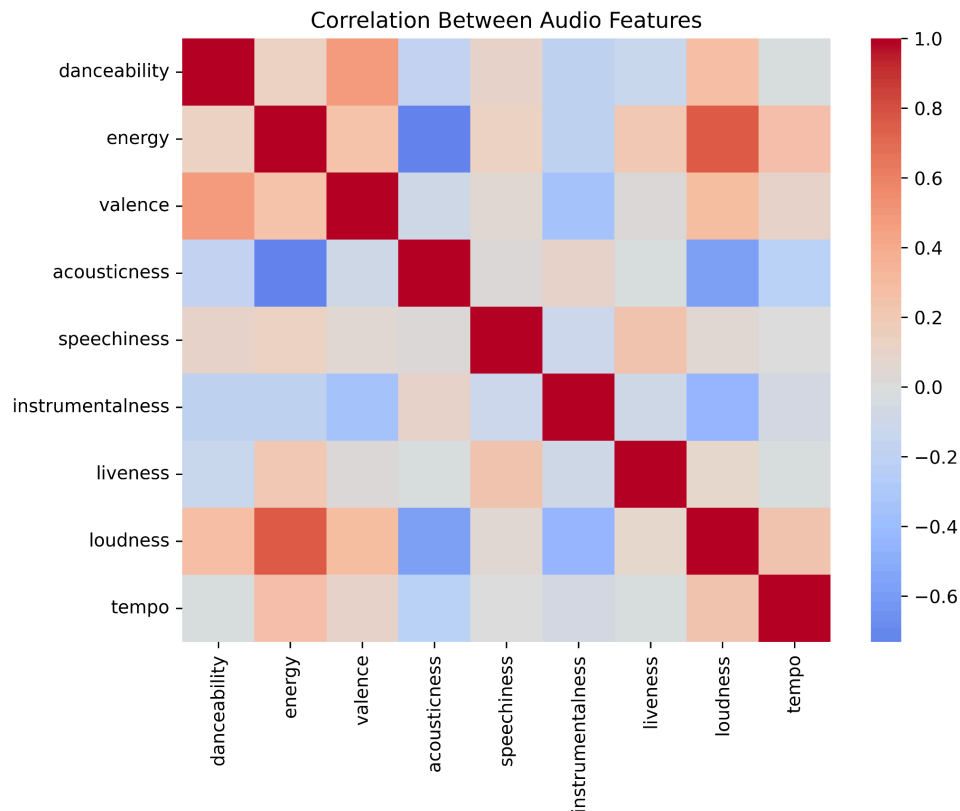


Figure 1: Correlation Between Audio Features

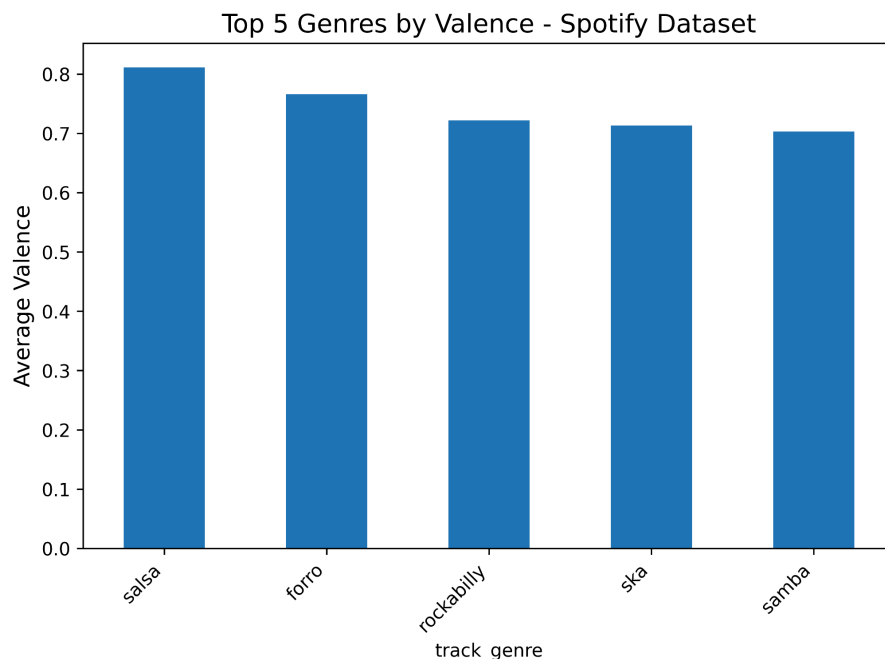


Figure 2a: Top 5 Genres by Average Valence

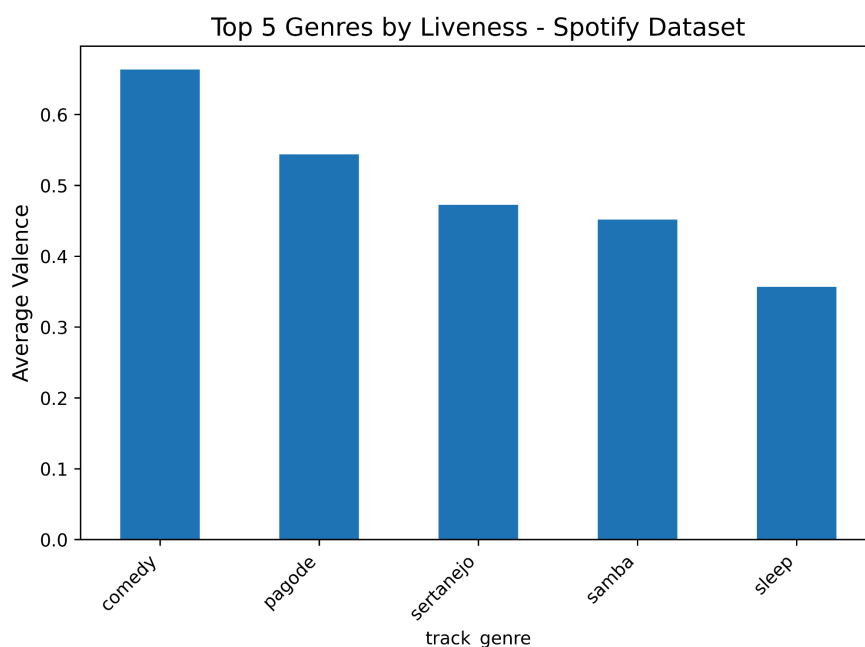


Figure 2b: Top 5 Genres by Average Liveness

Techniques Applied

Preprocessing and Data Preparation

Apart from modelling decisions, this was perhaps the most important step in developing the music recommendation system. Preprocessing decisions were challenging because of how artists and songs are named. Consider artists such as *A\$AP Rocky*, which have a dollar sign in their name. Or, consider songs

which have mandarin or non-English characters. Since the aim was for the user to enter a track and artist name at the front-end, and for the algorithm to generate recommendations based on that, it was critical to preprocess track and artists' text labels to support the intended infrastructure. Several preprocessing steps were taken to do so, including:

Text normalization diagnostics

Original text fields for track name, artist name, and album name were retained in raw form for display and use at the front-end. Cleaned versions of these features were created for internal matching. Cleaning involved lowercasing, removing punctuation and additional metadata in track names (e.g., live or remastered tags), normalizing non-English or non-Latin characters to English one, and handling symbol-based lettering (e.g., mapping "\$" to "s").

Deduplication

Exploratory diagnostics revealed that many songs appeared multiple times due to Spotify's album-based structure. For example, the same song by the same artist can appear across multiple albums, such as the original release, greatest hits, playlist compilations, etc. The dataset was deduplicated by retaining a single unit per song–artist pair, while selecting the most popular instance to represent each song.

Audio Feature Selection

The project only retained a core set of continuous audio features: danceability, energy, valence, acousticness, speechiness, instrumentality, liveness, loudness, and tempo. All features were standardized prior to modeling, since they were on different scales. Features such as key, mode, and time signature were excluded because they are either categorical or circular (such as in the case of Key), or did not have much variance in them (such as in the case of Time Signature).

Principal Component Analysis

As shown in Figure 1 above, some audio features showed substantial correlation. To address redundancy and reduce dimensionality, I applied Principal Component Analysis (PCA) after standardization. The cumulative explained variance plot showed that the first four principal components captured over 80% of total variance across audio features. However, for visual ease, PCA was reduced to two principal components only, which was then used both for visualization and as well as a precursor to clustering to improve computational efficiency. Figure 3 shows the distribution of the reduced audio feature space based on the two principal components.

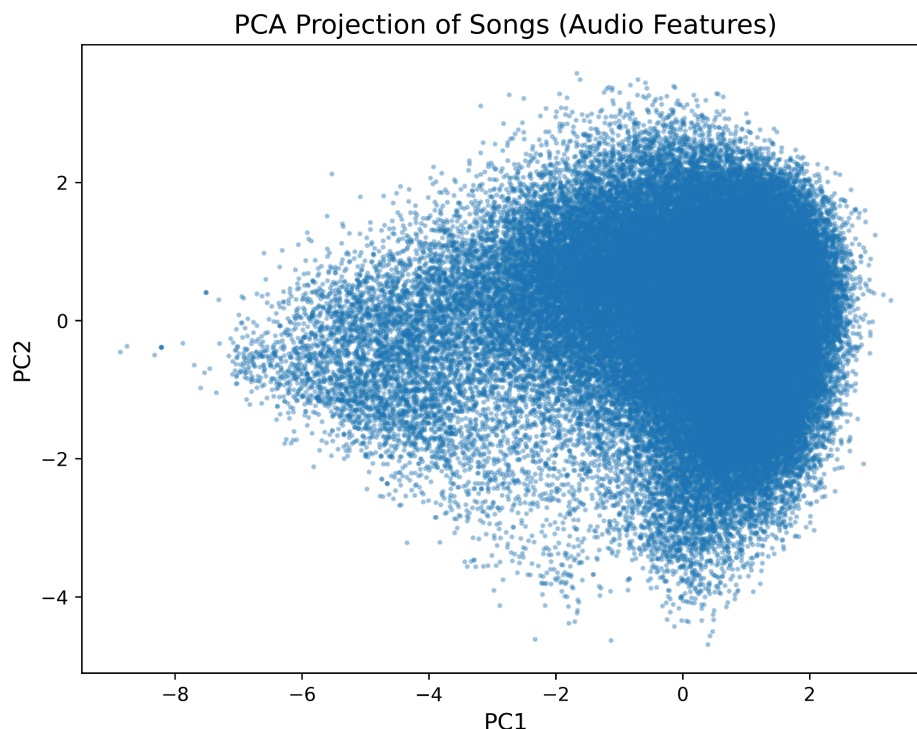


Figure 3: PCA Visualization - Spotify Dataset (Two Principal Components)

K-Means Clustering

To identify groups of musically similar songs, I applied K-means clustering to the PCA-transformed feature space. There were several reasons to choose K-Means as the clustering approach here. Since genres are not clearly partitional and overlap, it gave me the ability to experiment with the number of clusters. K-Means has also been used in developing algorithms based on audio features of songs in other projects as well (Schedl, Gómez, and Urbano, 2014). Multiple values of K were evaluated using inertia based elbow plots, silhouette scores, the cluster size with different values of K , and cluster cohesion. I also checked whether the clusters generated were interpretable or not.

Cosine Similarity and Recommendation Logic

After clustering, cosine similarity was used to generate recommendations and measure how similar two songs were in the audio features space. There were 5 songs recommended for three categories of recommendations mentioned below. Cosine similarity score was used to rank these recommendations.

1. **Similar songs by the same artist:** Recommending musically similar songs filtered by the same artist from the same cluster
2. **Other popular hits:** Recommending similar songs from the same cluster but not from the same artist. Filtered for high popularity using the Popularity feature
3. **Similar hidden gems:** Recommending similar songs from the same cluster but not from the same artist. Filtered for low popularity using the Popularity feature

The aim was to make a diverse and balanced set of recommendations to the user. I wanted to overcome the challenge associated with collaborative filtering, where mostly popular music is recommended. Using the Popularity feature as a filter ensured that the recommendations would include songs that were musically similar to the one entered by the user, but may not be as popular. The intent was to help with music discovery and help give exposure to artists who do not have popular acclaim.

Findings

Cluster Selection and Diagnostics

Based on the initial inertia-based elbow plots and silhouette scores, I deliberated between choosing either 4 or 5 clusters. In both cases, a small but distinct cluster emerged with extremely high Speechiness and Acousticness. Further investigation showed that this cluster was dominated by comedy, spoken word, children's content, and show tunes. It represented non-musical or speech-heavy tracks rather than songs.

I also found that 5 clusters did not meaningfully improve cluster separation. The clusters were also not as intuitive. In contrast, 4 clusters produced clearly identifiable and intuitive categories, including one spoken word cluster and three music based clusters with distinct audio profiles. The relative cluster size and cohesion was also more reasonable. For these reasons $K = 4$ were selected as the final clustering solution. The spoken-word cluster was retained but treated separately in recommendation logic.

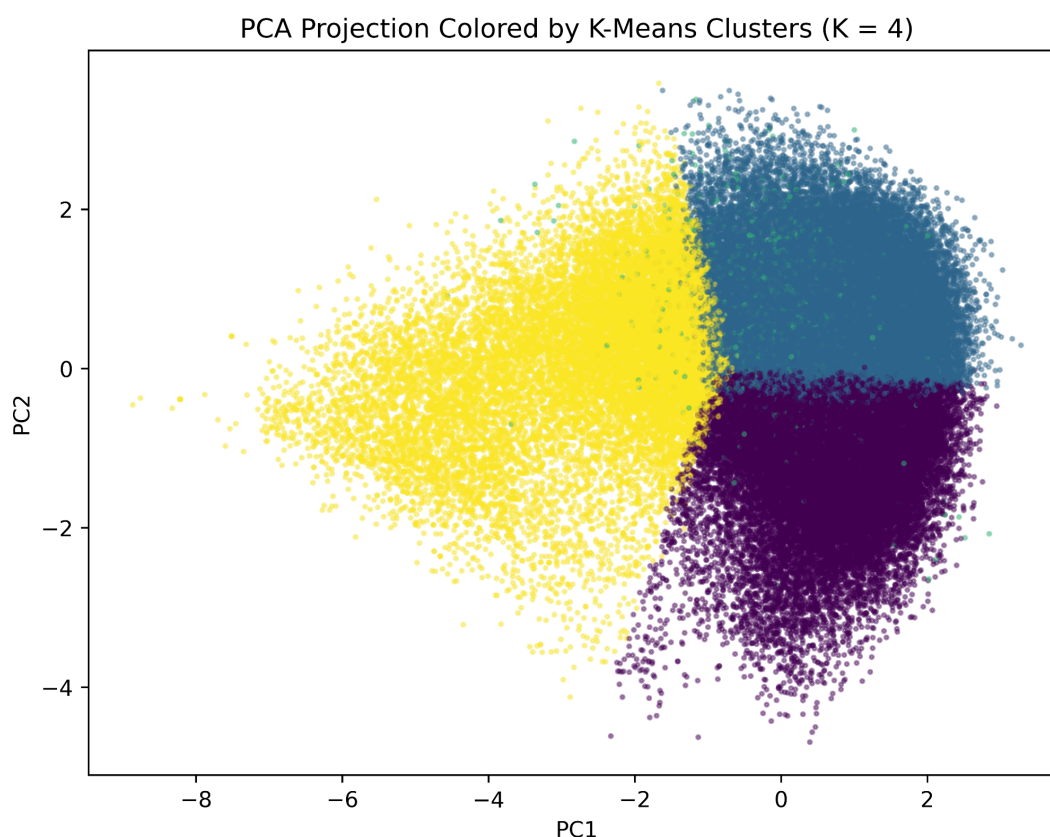


Figure 4: $K = 4$ Clusters Generated by K-Means: As seen one cluster (in green) is spread over other clusters; this is related to songs with high Speechiness and Acousticness

Using genre distributions and average audio features, the four clusters generated through K-Means could be summarized as:

1. **Cluster 0: High-energy music:** Loud, fast-paced, and energetic tracks. Dominated by metal, hardstyle, drum-and-bass, and similar genres.
2. **Cluster 1: Dance-oriented music:** High danceability and valence with moderate energy. Included pop, afrobeat, Latin, and club-oriented genres.
3. **Cluster 2: Spoken-word tracks:** Extremely high speechiness and acousticness. Included comedy, children's content, and show tunes.
4. **Cluster 3: Calm, ambient music:** Low energy and loudness with high acousticness. Included ambient, sleep, new-age, and romantic genres.

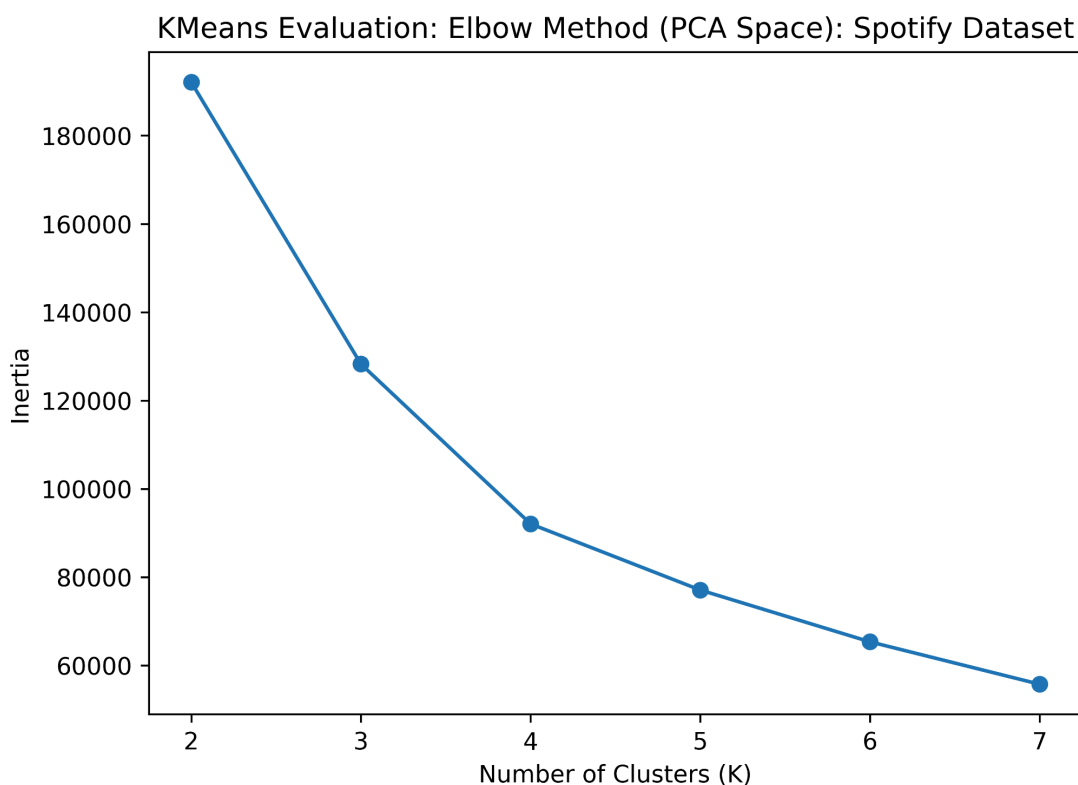


Figure 5: Elbow Plot of K-Means Inertia Computed on PCA-reduced Space: Can clearly see a bend in the elbow at $K = 4$

The results showed that Spotify's audio features contained enough information to meaningfully organize songs based on how they sound, even without genre labels or user listening data. K-means clustering, when applied to the PCA reduced space, produced four clusters that were both interpretable and relatively stable across diagnostics.

Three of the clusters corresponded to familiar genres. One cluster grouped together loud and fast-paced songs with high energy and tempo. This included aggressive or high-intensity music, such as metal and electronic genres. The second cluster captured dance-oriented tracks, characterized by high danceability and positive mood. This group included a mix of pop, Latin, afrobeat, and club music, reflecting genres

that were mostly focused on groove. The third cluster contained slower and more subdued music, with low energy and loudness but high acousticness. Genres associated with this cluster included ambient, sleep, and new-age music, which are often used for relaxation.

To assess whether this translated into useful recommendations, the recommendation logic was tested on individual songs. For a given input track, recommendations were drawn from the same cluster and ranked using cosine similarity. The recommendations generated showed similar characteristics, such as tempo, intensity, and overall mood. The use of popularity-based filters further allowed to balance familiarity with discovery by recommending both well-known songs and less popular tracks.

Overall, the findings suggested that a content-based recommendation system built solely on audio features can produce intuitive results. While it may not account for personal taste or user engagement, it could be a plausible alternative to collaborative filtering, and especially useful in music discovery.

Project Limitations

This project has several limitations. Firstly, there was no way to conclusively validate the quality of music recommendations. Music choices are very subjective, and without user interaction data, it is not possible to evaluate accuracy. For this reason, the system relied more on qualitative assessment of the recommendations made. To partially address this limitation, user feedback was incorporated into the Streamlit application developed for the music recommendation system, which is accessible [here](#). This allowed users to give feedback on whether recommendations felt useful, providing some user-level metrics.

Another limitation was the absence of personalization and exclusive reliance on audio features. The recommender did not account for individual listening habits or lyrical content, all of which can have an impact on the kind of music users listen to. Genre labels are also imperfect and sometimes inconsistent. Future work could therefore explore hybrid models that combine content based filtering combined with collaborative filtering. Such projects could also include lyrics along with user feedback to improve recommendation quality.

Bibliography

- Benito, S., Carr, A., & Sánchez-Arias, R. (n.d.). *Music data mining using audio features extracted from Spotify*. St. Thomas University. https://www.reisanar.com/files/poster_Benito_Carr_v5.pdf
- Kathavate, S. (2021). *Music recommendation system using content and collaborative filtering methods*. *International Journal of Engineering Research & Technology (IJERT)*, 10(02).
<https://www.ijert.org/music-recommendation-system-using-content-and-collaborative-filtering-methods>
- Kim, D.-M., Kim, K.-S., Park, K.-H., Lee, J.-H., & Lee, K. M. (2007). *A music recommendation system with a dynamic K-means clustering algorithm*. In *Proceedings of the Sixth International Conference on Machine Learning and Applications* (pp. 399–403). IEEE. <https://doi.org/10.1109/ICMLA.2007.97>
- Pandya, M. (2021). *Spotify tracks dataset*. Kaggle.
<https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset/data>
- Scaringella, N., Zoia, G., & Mlynek, D. (2006). *Automatic genre classification of music content: A survey*. *IEEE Signal Processing Magazine*, 23(2), 133–141.
<https://www.semanticscholar.org/paper/Automatic-genre-classification-of-music-content%3A-a-Scaringella-Zoia/7e7a9899a839eda239ebb001bc6b45002fc1a2b7>
- Schedl, M., Gómez, E., & Urbano, J. (2014). *Music information retrieval: Recent developments and applications*. *Foundations and Trends in Information Retrieval*, 8(2–3), 127–261.
<https://julian-urbano.info/files/publications/059-music-information-retrieval-recent-developments-applications.pdf>